

SRI VENKATESWARA UNIVERSITY COLLEGE OF ENGINEERING: TIRUPATI
DEPARTMENT OF COMPUTER SCIENCE ENGINEERING
Postgraduate Programme
M.Tech Scheme of Instruction for Choice Based Credit System
(With effect from 2023-24 admitted batch)

Total credits (2 year course): 68

SEMESTER I

Course Code	Course Title	L	T	P	Total	Credits	Sectional	End Semester	Total
COPC01	Mathematical Foundation of Computer Science	3	1	0	4	4	40	60	100
COPC02	Advanced Data Structures	3	1	0	4	4	40	60	100
COPC03	Advanced Operating Systems	3	1	0	4	4	40	60	100
COPE 1	Programme Elective-I	3			3	3	40	60	100
COPE11	Machine Learning								
COPE12	Data Preparation and Analysis								
COPE 2	Programme Elective-II	3			3	3	40	60	100
COPE21	Distributed Databases								
COPE22	Digital Forensics								
COCP 01	Core-I Lab:			3	3	1.5	40	60	100
COEP 01	Elective-I Lab			3	3	1.5	40	60	100
PGPC 01	Research Methodology and IPR	3	0	0	3	3	40	60	100
Total		18	3	6	27	24	320	480	800

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SEMESTER II

Course Code	Course Title	L	T	P	Total	Credits	Seasonal	End Semester	Total
COPC 04	Advance Algorithms	3	1	0	4	4	40	60	100
COPC 05	Compilers for HPC	3	1	0	4	4	40	60	100
COPE 3	Programme Elective-III	3	0	0	3	3	40	60	100
COPE31	Big Data Analytics								
COPE32	Recommender System								
COPE 4	Programme Elective-IV	3	0	0	3	3	40	60	100
COPE41	Computer Vision								
COPE42	Introduction to Intelligent Systems								
COCP02	Core-II Lab			3	3	1.5	40	60	100
COEP02	Elective-II Lab			3	3	1.5	40	60	100
PGPC02	Cyber Security	2	0	0	2	2	100	-	100
COMP 01	Mini Project with seminar	1		4	5	4	100	-	100
		15	2	10	27	23	440	360	800

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SEMESTER III

Course Code	Course Title	L	T	P	Credits	Sessional	End Semester	Total
COP1	Internship/Mini Project (Min 4 Weeks)	0			3	100		100
PGOE 11	MOOCs	0	0	0	3	100		100
PGOE111	Block chain Technology							
PGOE112	IoT							
PGOE113	Mobile Networks							
PGOE114	Network Security							
PGOE115	Any other in Current Technology							
COPD01	Dissertation-Phase-I			24	12	40	60	100
Total		0	0	24	18	240	60	300

- Courses selected under Moocs should not be same as Core and Electives

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SEMESTER IV

		L	T	P	Credits	Sessional	End Semester	Total
COPD 02	Dissertation-Phase-II			20	16	40	60	100
Total				20	16	100		100

SEMESTER I

PROGRAMME CORE

Course Code	COPC 01
Course Name	Mathematical Foundation of Computer Science
Credits	3
Pre-Requisites	Discrete Mathematics

Total Number of Lectures:48

COURSE OBJECTIVE
• To understand the mathematical fundamentals that is prerequisites for a variety of courses like Data mining, Network protocols, analysis of Web traffic, Computer security, Software engineering, Computer architecture, operating systems, distributed systems, Bioinformatics, Machine learning.
• To develop the understanding of the mathematical and logical basis to many modern techniques in information technology like machine learning, programming language design, and concurrency.
• To study various sampling and classification problems.

LECTURE WITH BREAKUP	NO. OF LECTURES
Unit 1 Probability mass, density, and cumulative distribution functions, Parametric families of distributions, Expected value, variance, conditional expectation, Applications of the univariate and multivariate Central Limit Theorem, Probabilistic inequalities, Markov chains	10
Unit 2 Random samples, sampling distributions of estimators, Methods of Moments and Maximum Likelihood,	9
Unit 3 Statistical inference, Introduction to multivariate statistical models: regression and classification problems, principal components analysis, The problem of overfitting model assessment.	8
Unit 4 Graph Theory: Isomorphism, Planar graphs, graph colouring, hamilton circuits and euler cycles. Permutations and Combinations with and without repetition. Specialized techniques to solve combinatorial enumeration problems	11
Unit 5 Computer science and engineering applications	10

Data mining, Network protocols, analysis of Web traffic, Computer security, Software engineering, Computer architecture, operating systems, distributed systems, Bioinformatics, Machine learning.	
COURSE OUTCOMES	
After completion of course, students would be able to:	
• To understand the basic notions of discrete and continuous probability.	
• To understand the methods of statistical inference, and the role that sampling distributions play in those methods.	
• To be able to perform correct and meaningful statistical analyses of simple to moderate complexity.	

References

1. John Vince, Foundation Mathematics for Computer Science, Springer.
2. K. Trivedi. Probability and Statistics with Reliability, Queuing, and Computer Science Applications. Wiley.
3. M. Mitzenmacher and E. Upfal. Probability and Computing: Randomized Algorithms and Probabilistic Analysis.
4. Alan Tucker, Applied Combinatorics, Wiley

Course Code	COPC 02
Course Name	Advanced Data Structures
Credits	3
Pre-Requisites	UG level course in Data Structures

Total Number of Lectures:48

COURSE OBJECTIVE
The student should be able to choose appropriate data structures, understand the ADT/libraries, and use it to design algorithms for a specific problem.
Students should be able to understand the necessary mathematical abstraction to solve problems.
To familiarize students with advanced paradigms and data structure used to solve algorithmic problems.
Student should be able to come up with analysis of efficiency and proofs of correctness.

LECTURE WITH BREAKUP	NO. OF LECTURES
Unit 1 Dictionaries: Definition, Dictionary Abstract Data Type, Implementation of Dictionaries. Hashing: Review of Hashing, Hash Function, Collision Resolution Techniques in Hashing, Separate Chaining, Open Addressing, Linear Probing, Quadratic Probing, Double Hashing, Rehashing, Extendible Hashing.	9
Unit 2 Skip Lists: Need for Randomizing Data Structures and Algorithms, Search and Update Operations on Skip Lists, Probabilistic Analysis of Skip Lists, Deterministic Skip Lists	8
Unit 3 Trees: Binary Search Trees, AVL Trees, Red Black Trees, 2-3 Trees, B-Trees, Splay Trees	9
Unit 4	12
Text Processing: String Operations, Brute-Force Pattern Matching, The Boyer-Moore Algorithm, The Knuth-Morris-Pratt Algorithm, Standard Tries, Compressed Tries, Suffix Tries, The Huffman Coding Algorithm, The Longest Common Subsequence Problem (LCS), Applying Dynamic Programming to the LCS Problem.	
Unit 5 Computational Geometry: One Dimensional Range Searching, Two Dimensional Range Searching, Constructing a Priority Search Tree, Searching a Priority Search Tree, Priority Range Trees, Quadrees, k-D Trees.	10

COURSE OUTCOMES
After completion of course, students would be able to:
Understand the implementation of symbol table using hashing techniques.
Develop and analyze algorithms for red-black trees, B-trees and Splay trees.
Develop algorithms for text processing applications.
Identify suitable data structures and develop algorithms for computational geometry problems.

References:

1. Mark Allen Weiss, Data Structures and Algorithm Analysis in C++, 2nd Edition, Pearson, 2004.
2. M T Goodrich, Roberto Tamassia, Algorithm Design, John Wiley, 2002.

Course Code	COPC 03
Course Name	Advanced Operating System
Credits	3
Pre-Requisites	Data Structure, Algorithms, Computer Architecture and Organization

Total Number of Lectures: 48

COURSE OBJECTIVE
The objective of the course is to provide introduction to operating system design and concept of process, process lifecycle and scheduling approaches.

LECTURE WITH BREAKUP	NO. OF LECTURES
Unit 1: Computer system and operating system overview, Operating system functions and design issues, Design approaches, Types of advanced operating systems.	10
Unit 2: Process abstraction, Process management, system calls, Threads, Symmetric multiprocessing and micro-kernels.	8

Unit 3: Scheduling: Uniprocessor, Multiprocessor and Real time systems, concurrency, classical problems, mechanisms for synchronization: semaphores, monitors, Process deadlock and deadlock handling strategies.	10
Unit 4: Memory management, Virtual memory concept, Virtual machines, I/O management, File and disk management, Operating system security.	9
Unit 5: Distributed Operating system: Architecture, Design issues, Distributed mutual exclusion, Distributed deadlock detection, shared memory, Distributed scheduling. Multiprocessor operating systems: architecture, operating system design issues, threads, process synchronization, process scheduling, memory management, reliability and fault tolerance.	11

COURSE OUTCOMES
After completion of course, students would be:
- Understanding advanced concepts in operating systems. - Learning principles of Distributed and multiprocessor operating systems

References:

1. Advanced concept in operating system: M. Singhal, N.G. Shivratri
2. Operating system internal and design principles: William Stallings

PROGRAMME ELECTIVE-I Total Number of Lectures:48

PROGRAMME ELECTIVE - Total Number of Lectures: 48	
Course Code	COPE 11
Course Name	Machine learning
Credits	3
Pre-Requisites	
COURSE OBJECTIVE	
To learn the concept of how to learn patterns and concepts from data without being explicitly programmed in various IOT nodes.	
To design and analyse various machine learning algorithms and techniques with a modern outlook focusing on recent advances.	
Explore supervised and unsupervised learning paradigms of machine learning.	
To explore Deep learning technique and various feature extraction strategies.	
LECTURE WITH BREAKUP	NO. OF LECTURES
Unit 1: Supervised Learning (Regression/Classification) Basic methods: Distance-based methods, Nearest-Neighbours, Decision Trees, Naive Bayes	10
- Linear models: Linear Regression, Logistic Regression, Generalized Linear Models - Support Vector Machines, Nonlinearity and Kernel Methods - Beyond Binary Classification: Multi-class/Structured Outputs, Ranking	
Unit 2: Unsupervised Learning - Clustering: K-means/Kernel K-means - Dimensionality Reduction: PCA and kernel PCA - Matrix Factorization and Matrix Completion - Generative Models (mixture models and latent factor models)	9
Unit 3 Evaluating Machine Learning algorithms and Model Selection, Introduction to Statistical Learning Theory, Ensemble Methods (Boosting, Bagging, Random Forests)	9
Unit 4 Sparse Modeling and Estimation, Modeling Sequence/Time-Series Data, Deep Learning and Feature Representation Learning	9
Unit 5 Scalable Machine Learning (Online and Distributed Learning) A selection from some other advanced topics, e.g., Semi-supervised Learning, Active Learning, Reinforcement Learning, Inference in Graphical Models, Introduction to Bayesian Learning and Inference	9
COURSE OUTCOMES	
After completion of course, students would be able to:	
Extract features that can be used for a particular machine learning approach in various IOT applications.	
To compare and contrast pros and cons of various machine learning techniques and to get an insight of when to apply a particular machine learning approach.	
To mathematically analyse various machine learning approaches and paradigms.	

References:1 Kevin Murphy, Machine Learning: A Probabilistic Perspective, MIT Press, 2012

- 2 Trevor Hastie, Robert Tibshirani, Jerome Friedman, The Elements of Statistical Learning, Springer 2009 (freely available online)
- 3 Christopher Bishop, Pattern Recognition and Machine Learning, Springer, 2007.

Course Code	COPE12
Course Name	Data Preparation and Analysis
Credits	3
Pre-Requisites	

Total Number of Lectures: 48

COURSE OBJECTIVE
To prepare the data for analysis and develop meaningful Data Visualizations

LECTURE WITH BREAKUP	NO. OF LECTURES
Unit1: Data Gathering and Preparation: Data formats, parsing and transformation, Scalability and real-time issues	9
Unit2: Data Cleaning: Consistency checking, Heterogeneous and missing data, Data Transformation and segmentation	11
Unit3: Exploratory Analysis: Descriptive and comparative statistics, Clustering and association, Hypothesis generation	13
Unit4: Visualization: Designing visualizations, Time series, Geolocated data, Correlations and connections, Hierarchies and networks, interactivity	10
Unit5: Current trends	5

COURSE OUTCOMES
After completion of course, students would be:
Able to extract the data for performing the Analysis.

References:

1. Making sense of Data : A practical Guide to Exploratory Data Analysis and Data Mining, by Glenn J. Myatt

PROGRAMME ELECTIVE-II

Course Code	COPE 21
Course Name	Distributed Database
Credits	3
Pre-Requisites	

Total Number of Lectures: 48

COURSE OBJECTIVE
The objective of course is to provide insight to distributed database, normalization techniques and integrity rules. It also includes parallel database systems along with object oriented models.

LECTURE WITH BREAKUP	NO. OF LECTURES
Unit 1: Introduction: Distributed Data processing, Distributed database system (DDBMS), Promises of DDBMSs, Complicating factors and Problem areas in DDBMSs, Overview Of Relational DBMS Relational Database concepts, Normalization, Integrity rules, Relational Data Languages, Relational DBMS.	11
Unit 2: Distributed DBMS Architecture: DBMS Standardization, Architectural models for Distributed DBMS, Distributed DBMS Architecture. Distributed Database Design: Alternative design Strategies, Distribution design issues, Fragmentation, Allocation. Semantic Data Control: View Management, Data security, Semantic	10
Integrity Control.	
Unit 3: Overview of Query Processing: Query processing problem, Objectives of Query Processing, Complexity of Relational Algebra operations, characterization of Query processors, Layers of Query Processing. Introduction to Transaction Management: Definition of Transaction, Properties of transaction, types of transaction. Distributed Concurrency Control: Serializability theory, Taxonomy of concurrency control mechanisms, locking bases concurrency control algorithms.	9
Unit 4: Parallel Database Systems: Database servers, Parallel architecture, Parallel DBMS techniques, Parallel execution problems, Parallel execution for hierarchical architecture.	10
Unit 5: Distributed Object Database Management systems: Fundamental Object concepts and Object models, Object distribution design. Architectural issues, Object management, Distributed object storage, Object query processing. Transaction management. Database Interoperability: Database Integration, Query processing.	8

COURSE OUTCOMES
After completion of course, students would be:

- Able to understand relational database management systems, normalization to make efficient retrieval from database and query.

References:

- 1.Principles of Distributed Database Systems, Second Edition, M. Tamer Ozsu Patrick Valduriez
- 2.Distributed Databases principles and systems, Stefano Ceri, Giuseppe Pelagatti, Tata McGraw Hill.

Course Code	COPE 22
Course Name	Digital Forensics
Credits	3
Pre-Requisites	Cybercrime and Information Warfare, Computer Networks

Total Number of Lectures: 48

COURSE OBJECTIVE	
• Provides an in-depth study of the rapidly changing and fascinating field of computer forensics.	
• Combines both the technical expertise and the knowledge required to investigate, detect and prevent digital crimes.	
• Knowledge on digital forensics legislations, digital crime, forensics processes and procedures, data acquisition and validation, e-discovery tools	
• E-evidence collection and preservation, investigating operating systems and file systems, network forensics, art of steganography and mobile device forensics	
LECTURE WITH BREAKUP	NO. OF LECTURES
Unit 1: Digital Forensics Science: Forensics science, computer forensics, and digital forensics. Computer Crime: Criminalistics as it relates to the investigative process, analysis of cyber-criminalistics area, holistic approach to cyber-forensics	9
Unit 2: Cyber Crime Scene Analysis: Discuss the various court orders etc., methods to search and seizure electronic evidence, retrieved and un-retrieved communications, Discuss the importance of understanding what court documents would be required for a criminal investigation.	10
Unit 3: Evidence Management & Presentation: Create and manage shared folders using operating system, importance of the forensic mindset, define the workload of law enforcement, Explain what the normal case would look like, Define who should be notified of a crime, parts of gathering evidence, Define and apply probable cause.	9
Unit 4: Computer Forensics: Prepare a case, Begin an investigation, Understand computer forensics workstations and software, Conduct an investigation, Complete a case, Critique a case, Network Forensics: open-source security tools for network forensic analysis, requirements for preservation of network data.	10
Unit 5: Mobile Forensics: mobile forensics techniques, mobile forensics tools.	10

Legal Aspects of Digital Forensics: IT Act 2000, amendment of IT Act 2008.	
COURSE OUTCOMES	
After completion of course, students would be able to:	
• Understand relevant legislation and codes of ethics	
• Computer forensics and digital detective and various processes, policies and procedures	
• E-discovery, guidelines and standards, E-evidence, tools and environment.	
• Email and web forensics and network forensics	

References:1 John Sammons, The Basics of Digital Forensics, Elsevier

2 John Vacca, Computer Forensics: Computer Crime Scene Investigation, Laxmi Publications

1. gn, John Willey, 2002.

Course Code	PGPC 01
Course Name	Research Methodology and IPR
Credits	3

Research Methodology and IPR

Teaching Scheme

Lectures: 1hrs/week

nation, it is needless to emphasis the need of information about Intellectual Property Right to be promoted among students in general & engineering in particular.

- Understand that IPR protection provides an incentive to inventors for further research work and investment in R & D, which leads to creation of new and better products, and in turn brings about, economic growth and social benefits.

Syllabus Contents:

Unit 1: Meaning of research problem, Sources of research problem, Criteria Characteristics of a good research problem, Errors in selecting a research problem, Scope and objectives of research problem.

Approaches of investigation of solutions for research problem, data collection, analysis, interpretation, Necessary instrumentations

Unit 2: Effective literature studies approaches, analysis

Plagiarism, Research ethics,

Unit 3: Effective technical writing, how to write report, Paper

Developing a Research Proposal, Format of research proposal, a presentation and assessment by a review committee

Unit 4: Nature of Intellectual Property: Patents, Designs, Trade and Copyright. Process of Patenting and Development: technological research, innovation, patenting, development. International Scenario: International cooperation on Intellectual Property. Procedure for grants of patents, Patenting under PCT.

Unit 5: Patent Rights: Scope of Patent Rights. Licensing and transfer of technology. Patent information and databases. Geographical Indications.

Unit 6: New Developments in IPR: Administration of Patent System. New developments in IPR; IPR of Biological Systems, Computer Software etc. Traditional knowledge Case Studies, IPR and IITs.

References:

- Stuart Melville and Wayne Goddard, “Research methodology: an introduction for science & engineering students”
- Wayne Goddard and Stuart Melville, “Research Methodology: An Introduction”
- Ranjit Kumar, 2 nd Edition , “Research Methodology: A Step by Step Guide for beginners”
- Halbert, “Resisting Intellectual Property”, Taylor & Francis Ltd ,2007.
- Mayall , “Industrial Design”, McGraw Hill, 1992.
- Niebel , “Product Design”, McGraw Hill, 1974.
- Asimov , “Introduction to Design”, Prentice Hall, 1962.
- Robert P. Merges, Peter S. Menell, Mark A. Lemley, “ Intellectual Property in New Technological Age”, 2016.
- T. Ramappa, “Intellectual Property Rights Under WTO”, S. Chand, 2008

SEMESTER II

PROGRAMME CORE

Course Code	COPC 04
Course Name	Advanced Algorithms
Credits	3
Pre-Requisites	UG level course in Algorithm Design and Analysis

Total Number of Lectures:48

COURSE OBJECTIVE
● Introduce students to the advanced methods of designing and analyzing algorithms. ● The student should be able to choose appropriate algorithms and use it for a specific problem. ● To familiarize students with basic paradigms and data structures used to solve advanced algorithmic problems. ● Students should be able to understand different classes of problems concerning their computation difficulties. ● To introduce the students to recent developments in the area of algorithmic design.

LECTURE WITH BREAKUP	NO. OF LECTURES
Unit1 Sorting: Review of various sorting algorithms, topological sorting Graph: Definitions and Elementary Algorithms: Shortest path by BFS, shortest path in edge-weighted case (Dijkstra's), depth-first search and computation of strongly connected components, emphasis on correctness proof of the algorithm and time/space analysis, example of amortized analysis.	9
Unit 2 Matroids: Introduction to greedy paradigm, algorithm to compute a maximum weight maximal independent set. Application to MST. Graph Matching: Algorithm to compute maximum matching. Characterization of maximum matching by augmenting paths, Edmond's Blossom algorithm to compute augmenting path.	10

Unit 3 Flow-Networks: Maxflow-mincut theorem, Ford-Fulkerson Method to compute maximum flow, Edmond-Karp maximum-flow algorithm. Matrix Computations: Strassen's algorithm and introduction to divide and conquer paradigm, inverse of a triangular matrix, relation between the time complexities of basic matrix operations, LUP-decomposition.	9
Unit 4 Shortest Path in Graphs: Floyd-Warshall algorithm and introduction to dynamic programming paradigm. More examples of dynamic programming. Modulo Representation of integers/polynomials: Chinese Remainder Theorem, Conversion between base-representation and modulo-representation. Extension to polynomials. Application: Interpolation problem. Discrete Fourier Transform (DFT): In complex field, DFT in modulo ring. Fast Fourier Transform algorithm. Schonhage-Strassen Integer Multiplication algorithm	10
Unit 5 Linear Programming: Geometry of the feasibility region and Simplex algorithm NP-completeness: Examples, proof of NP-hardness and NP-completeness. One or more of the following topics based on time and interest Approximation algorithms, Randomized Algorithms, Interior Point Method, Advanced Number Theoretic Algorithm	10

COURSE OUTCOMES
After completion of course, students would be able to:
Analyze the complexity/performance of different algorithms.
Determine the appropriate data structure for solving a particular set of problems.
Categorize the different problems in various classes according to their complexity.
Students should have an insight of recent activities in the field of the advanced data structure.

References:

1. "Introduction to Algorithms" by Cormen, Leiserson, Rivest, Stein.
2. "The Design and Analysis of Computer Algorithms" by Aho, Hopcroft, Ullman.
3. "Algorithm Design" by Kleinberg and Tardos.

Course Code	COPE 05
Course Name	Compiler for HPC
Credits	3
Pre-Requisites	Data Structure, Compiler Design, Theory of Computation

Total Number of Lectures: 48

COURSE OBJECTIVE
The objective of this course is to introduce structure of compilers and high performance compiler design for students. Concepts of cache coherence and parallel loops in compilers are included.

LECTURE WITH BREAKUP	NO. OF LECTURES
Unit1: High Performance Systems, Structure of a Compiler, Programming Language Features, Languages for High Performance.	9
Unit2: Data Dependence: Data Dependence in Loops, Data Dependence in Conditionals, Data Dependence in Parallel Loops, Program Dependence Graph. Scalar Analysis with Factored Use-Def Chains: Constructing Factored Use-Def Chains, FUD Chains for Arrays, Induction Variables Using FUD Chains, Constant Propagation with FUD Chains, Data Dependence for Scalars. Data Dependence Analysis for Arrays.	10
Unit3: Array Region Analysis, Pointer Analysis, I/O Dependence, Procedure Calls, Inter-procedural Analysis. Loop Restructuring: Simple Transformations, Loop Fusion, Loop Fission, Loop Reversal, Loop Interchanging, Loop Skewing, Linear Loop Transformations, Strip-Mining, Loop Tiling, Other Loop Transformations, and Inter-procedural Transformations. Optimizing for Locality: Single Reference to Each Array, Multiple References, General Tiling, Fission and Fusion for Locality.	10
Unit4: Concurrency Analysis: Concurrency from Sequential Loops, Concurrency from Parallel Loops, Nested Loops, Round off Error, Exceptions and Debuggers. Vector Analysis: Vector Code, Vector Code from Sequential Loops, Vector Code from For all Loops, Nested Loops, Round off Error, Exceptions, and Debuggers, Multi-vector Computers.	10
Unit5: Message-Passing Machines: SIMD Machines, MIMD Machines, Data Layout, Parallel Code for Array Assignment, Remote Data Access, Automatic Data Layout, Multiple Array Assignments, Other Topics.	10
Scalable Shared-Memory Machines: Global Cache Coherence, Local Cache Coherence, Latency Tolerant Machines.	

COURSE OUTCOMES
After completion of course, students would be:

- Familiar with the structure of compiler.
- Parallel loops, data dependency and exception handling and debugging in compiler.

References:1. Michael Wolfe, High-Performance Compilers for Parallel Computing, Pearson

PROGRAMME ELECTIVE- III

Course Code	COPE31
Course Name	Big Data Analytics
Credits	3
Pre-Requisites	Data Structure, Computer Architecture and Organization

Total Number of Lectures: 48

COURSE OBJECTIVE
• Understand big data for business intelligence. Learn business case studies for big data analytics. Understand nosql big data management. Perform map-reduce analytics using Hadoop and related tools

LECTURE WITH BREAKUP	NO. OF LECTURES
Unit 1: What is big data, why big data, convergence of key trends, unstructured data, industry examples of big data, web analytics, big data and marketing, fraud and big data, risk and big data, credit risk management, big data and algorithmic trading, big data and healthcare, big data in medicine, advertising and big data, big data technologies, introduction to Hadoop, open source technologies, cloud and big data, mobile business intelligence, Crowd sourcing analytics, inter and trans firewall analytics.	8
Unit 2: Introduction to NoSQL, aggregate data models, aggregates, key-value and document data models, relationships, graph databases, schemaless databases, materialized views, distribution models, sharding, master-slave replication, peer- peer replication, sharding and replication, consistency, relaxing consistency, version stamps, map-reduce, partitioning and combining, composing map-reduce calculations.	8
Unit 3: Data format, analyzing data with Hadoop, scaling out, Hadoop streaming, Hadoop pipes, design of Hadoop distributed file system (HDFS), HDFS concepts, Java interface, data flow, Hadoop I/O, data integrity, compression, serialization, Avro, file-based data structures	9
Unit 4: MapReduce workflows, unit tests with MRUnit, test data and local tests, anatomy of MapReduce job run, classic Map-reduce, YARN, failures in classic Map-reduce and YARN, job scheduling, shuffle and sort, task execution, MapReduce types, input formats, output formats	10
Unit 5:	13
Hbase, data model and implementations, Hbase clients, Hbase examples, praxis.Cassandra, Cassandra data model, Cassandra examples, Cassandra clients, Hadoop integration. Pig, Grunt, pig data model, Pig Latin, developing and testing Pig Latin scripts. Hive, data types and file formats, HiveQL data definition, HiveQL data manipulation, HiveQL queries.	

COURSE OUTCOMES
After completion of course, students would be:
● Describe big data and use cases from selected business domains ● Explain NoSQL big data management ● Install, configure, and run Hadoop and HDFS ● Perform map-reduce analytics using Hadoop ● Use Hadoop related tools such as HBase, Cassandra, Pig, and Hive for big data analytics

References:

1. Michael Minelli, Michelle Chambers, and AmbigaDhiraj, "Big Data, Big Analytics: Emerging Business Intelligence and Analytic Trends for Today's Businesses", Wiley, 2013.
2. P. J. Sadalage and M. Fowler, "NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence", Addison-Wesley Professional, 2012.
3. Tom White, "Hadoop: The Definitive Guide", Third Edition, O'Reilley, 2012.
4. Eric Sammer, "Hadoop Operations", O'Reilley, 2012.
5. E. Capriolo, D. Wampler, and J. Rutherglen, "Programming Hive", O'Reilley, 2012.
6. Lars George, "HBase: The Definitive Guide", O'Reilley, 2011.
7. Eben Hewitt, "Cassandra: The Definitive Guide", O'Reilley, 2010.
8. Alan Gates, "Programming Pig", O'Reilley, 2011.

Course Code	COPE32
Course Name	Recommender System
Credits	3
Prerequisites	

Total Number of Lectures: 48

COURSE OBJECTIVE
☐ To learn techniques for making recommendations, including non-personalized, content-based, and collaborative filtering ☐ To automate a variety of choice-making strategies with the goal of providing affordable, personal, and high-quality recommendations

LECTURE WITH BREAKUP	NO. OF LECTURES
Unit 1: Introduction: Overview of Information Retrieval, Retrieval Models, Search and Filtering Techniques: Relevance Feedback, User Profiles, Recommender system functions, Matrix operations, covariance matrices, Understanding ratings, Applications of recommendation systems, Issues with recommender system.	9
Unit 2: Content-based Filtering: High level architecture of content-based systems, Advantages and drawbacks of content based filtering, Item profiles, Discovering features of documents, pre-processing and feature extraction, Obtaining item features from tags, Methods for learning user profiles, Similarity based retrieval, Classification algorithms.	8
Unit 3: Collaborative Filtering: User-based recommendation, Item-based recommendation, Model based approaches, Matrix factorization, Attacks on collaborative recommender systems.	9
Unit 4: Hybrid approaches: Opportunities for hybridization, Monolithic hybridization design: Feature combination, Feature augmentation, Parallelized hybridization design: Weighted, Switching, Mixed, Pipelined hybridization design: Cascade Meta-level, Limitations of hybridization strategies	8
Unit 5: Evaluating Recommender System: Introduction, General properties of evaluation research, Evaluation designs: Accuracy, Coverage, confidence, novelty, diversity, scalability, serendipity, Evaluation on historical datasets, Offline evaluations. Types of Recommender Systems: Recommender systems in personalized web search, knowledge-based recommender system, Social tagging recommender systems, Trust-centric recommendations, Group recommender systems.	14

COURSE OUTCOMES
After completion of course, students would be able to:

- Design recommendation system for a particular application domain.
- Evaluate recommender systems on the basis of metrics such as accuracy, rank accuracy, diversity, product coverage, and serendipity

References:

1. Jannach D., Zanker M. and FelFering A., Recommender Systems: An Introduction, Cambridge University Press (2011), 1st ed.
2. Charu C. Aggarwal, Recommender Systems: The Textbook, Springer (2016), 1st ed.
3. Ricci F., Rokach L., Shapira D., Kantor B.P., Recommender Systems Handbook, Springer(2011), 1st ed.
4. Manouselis N., Drachsler H., Verbert K., Duval E., Recommender Systems For Learning, Springer (2013), 1st ed.

PROGRAMME ELECTIVE-IV

Course Code	COPE41
Course Name	Computer Vision
Credits	3
Pre-Requisites	Linear algebra, vector calculus, Data structures and Programming.

Total Number of Lectures: 48

COURSE OBJECTIVE
● Be familiar with both the theoretical and practical aspects of computing with images.
● Have described the foundation of image formation, measurement, and analysis.
● Understand the geometric relationships between 2D images and the 3D world.
● Grasp the principles of state-of-the-art deep neural networks.

LECTURE WITH BREAKUP	NO. OF LECTURES
Unit 1: Overview, computer imaging systems, lenses, Image formation and sensing, Image analysis, pre-processing and Binary image analysis	10
Unit 2: Edge detection, Edge detection performance, Hough transform, corner detection	10
Unit 3: Segmentation, Morphological filtering, Fourier transform	9
Unit 4: Feature extraction, shape, histogram, color, spectral, texture, using CVIPtools, Feature analysis, feature vectors, distance /similarity measures, data pre-processing	10
Unit 5: Pattern Analysis: Clustering: K-Means, K-Medoids, Mixture of Gaussians Classification: Discriminant Function, Supervised, Un-supervised, Semi-supervised Classifiers: Bayes, KNN, ANN models; Dimensionality Reduction: PCA, LDA, ICA, and Non-parametric methods.	9

COURSE OUTCOMES
After completion of course, students would be able to:
● Developed the practical skills necessary to build computer vision applications.
● To have gained exposure to object and scene recognition and categorization from images.

References:

1. Computer Vision: Algorithms and Applications by Richard Szeliski.
2. Deep Learning, by Goodfellow, Bengio, and Courville.
3. Dictionary of Computer Vision and Image Processing, by Fisher et al.

Course Code	COPE42
Course Name	Introduction to Intelligent Systems
Credits	3
Pre-Requisites	Data Structures and Data Management or Data Structures

Total Number of Lectures: 48

COURSE OBJECTIVE	
The aim of the course is to introduce to the field of Artificial Intelligence (AI) with emphasis on its use to solve real world problems for which solutions are difficult to express using the traditional algorithmic approach. It explores the essential theory behind methodologies for developing systems that demonstrate intelligent behaviour including dealing with uncertainty, learning from experience and following problem solving strategies found in nature.	
LECTURE WITH BREAKUP	NO. OF LECTURES
Unit 1: Biological foundations to intelligent systems I: Artificial neural networks, Back-propagation networks, Radial basis function networks, and recurrent networks.	9
Unit 2: Biological foundations to intelligent systems II: Fuzzy logic, knowledge Representation and inference mechanism, genetic algorithm, and fuzzy neural networks.	8
Unit 3: Search Methods Basic concepts of graph and tree search. Three simple search methods: breadth-first search, depth-first search, iterative deepening search. Heuristic search methods: best-first search, admissible evaluation functions, hill-climbing search. Optimisation and search such as stochastic annealing and genetic algorithm.	10
Unit 4: Knowledge representation and logical inference Issues in knowledge representation. Structured representation, such as frames, and scripts, semantic networks and conceptual graphs. Formal logic and logical inference. Knowledge-based systems structures, its basic components. Ideas of Blackboard architectures.	9
Unit 5: Reasoning under uncertainty and Learning Techniques on uncertainty reasoning such as Bayesian reasoning, Certainty factors and Dempster-Shafer Theory of Evidential reasoning, A study of different learning and evolutionary algorithms, such as statistical learning and induction learning.	7

COURSE OUTCOMES
After completion of course, students would be:
Able to Demonstrate knowledge of the fundamental principles of intelligent systems and would be able to analyse and compare the relative merits of a variety of AI problem solving techniques.

References:

1. Luger G.F. and Stubblefield W.A. (2008). Artificial Intelligence: Structures and strategies for Complex Problem Solving. Addison Wesley, 6th edition.
2. Russell S. and Norvig P. (2009). Artificial Intelligence: A Modern Approach. Prentice-

Hall, 3rd edition.

Course Code	PGPC 02
Course Name	Cyber Security
Credits	2
Pre-Requisites	Information Communication Technology

Total Number of Lectures:48

COURSE OBJECTIVE
(a) Learn the foundations of Cyber security and threat landscape. (b) To equip students with the technical knowledge and skills needed to protect and defend against cyber threats. (c) To develop skills in students that can help them plan, implement, and monitor cyber security mechanisms to ensure the protection of information technology assets. (d) To expose students to governance, regulatory, legal, economic, environmental, social and ethical contexts of cyber security. (e) To expose students to responsible use of online social media networks. (f)) To systematically educate the necessity to understand the impact of cyber crimes and threats with solutions in a global and societal context.

LECTURE WITH BREAKUP	NO. OF LECTURES
Unit 1: Cyber security increasing threat landscape, Cyber security terminologies- Cyberspace, attack, attack vector, attack surface, threat, risk, vulnerability, exploit, exploitation, hacker., Non-state actors, Cyber terrorism, Protection of end user machine, Critical IT and National Critical Infrastructure, Cyberwarfare, Case Studies.	10
Unit 2: Cyber crimes targeting Computer systems and Mobiles- data diddling attacks, spyware, logic bombs, DoS, DDoS, APTs, virus, Trojans, ransomware, data breach., Online scams and frauds- email scams, Phishing, Vishing, Smishing, Online job fraud, Online sextortion, Debit/ credit card fraud, Online payment fraud, Cyberbullying, website defacement, Cyber- squatting, Pharming, Cyber espionage, Cryptojacking, Darknet- illegal trades, drug trafficking, human trafficking., Social Media Scams & Frauds- impersonation, identity theft, job scams, misinformation, fake news cyber crime against persons - cyber grooming, child pornography, cyber stalking., Social Engineering attacks, Cyber Police stations, Crime reporting procedure, Case studies.	10
Unit 3: Cyber crime and legal landscape around the world, IT Act,2000 and its amendments. Limitations of IT Act, 2000. Cyber crime and punishments, Cyber Laws and Legal and ethical aspects related to new technologies- AI/ML, IoT, Blockchain, Darknet and Social media, Cyber Laws of other countries, Case Studies.	10

Unit 4: Defining data, meta-data, big data, non- personal data. Data protection, Data privacy and data security, Personal Data Protection Bill and its compliance, Data protection principles, Big data security issues and challenges, Data protection regulations of other countries- General Data Protection Regulations(GDPR),2016 Personal Information Protection and Electronic Documents Act (PIPEDA)., Social media- data privacy and security issues.	8
Unit 5: Cyber security Plan- cyber security policy, cyber crises management plan., Business continuity, Risk assessment, Types of security controls and their goals, Cyber security audit and compliance, National cyber security policy and strategy.	10

COURSE OUTCOMES

On completion of the course the student should be able to

- (a) Understand the cyber security threat landscape.
- (b) Develop a deeper understanding and familiarity with various types of cyberattacks, cyber crimes, vulnerabilities and remedies thereto.
- (c) Analyse and evaluate existing legal framework and laws on cyber security.
- (d) Analyse and evaluate the digital payment system security and remedial measures against digital payment frauds.
- (g) Analyse and evaluate the importance of personal data its privacy and security.
- (h)) Analyse and evaluate the security aspects of social media platforms and ethical aspects associated with use of social media.
- (i) Analyse and evaluate the cyber security risks.
- (j) Based on the Risk assessment, plan suitable security controls , audit and compliance.
- (k) Evaluate and communicate the human role in security systems with an emphasis o n ethics, social engineering vulnerabilities and training.
- (l) Increase awareness about cyber-attack vectors and safety against cyber-frauds.
- (m) Take measures for self-cyber-protection as well as societal cyber-protection.

References:

1. Cyber Security Understanding Cyber Crimes, Computer Forensics and Legal Perspectives by Sumit Belapure and Nina Godbole, Wiley India Pvt. Ltd.
2. Information Warfare and Security by Dorothy F. Denning, Addison Wesley.
3. Security in the Digital Age: Social Media Security Threats and Vulnerabilities by Henry A. Oliver, Create Space Independent Publishing Platform.
4. Data Privacy Principles and Practice by Natraj Venkataramanan and Ashwin Shriram, CRC Press.
5. Information Security Governance, Guidance for Information Security Managers by

W. KragBrothy, 1st Edition, Wiley Publication